

ROGER CREUS CASTANYER

PHD STUDENT



INFORMATION

- 📞 438-528-9561
- ✉ creus99@protonmail.com
- 🌐 roger-creus.github.io
- 📍 Montreal, Quebec, Canada
- 📅 Expected PhD Graduation: Aug. 2028
- 🔗 Python / PyTorch / JAX / C++ / C#

EDUCATION



Research MSc in Computer Science & AI

Mila Quebec AI Institute and University of Montreal

Sep 2022 - Sep 2024

Member of the Robotics and Embodied AI lab (REAL), supervised by Professor Glen Berseth.

GPA: 4.225/4.3



BSc with Honours in Data Science and Engineering

Polytechnic University of Catalonia

Sep. 2017 - June. 2021

GPA: 7.18/10.0

LANGUAGES

- 🇬🇧 English
- 🇪🇸 Spanish
- 🇪🇦 Catalan
- 🇫🇷 French

PROFILE

I am from Barcelona (Spain) and live in Montreal (Canada) as I pursue my PhD at Mila Quebec AI Institute and the University of Montreal, supervised by Professors *Pablo Samuel Castro* and *Glen Berseth*.

My research focuses on developing **intelligent agentic systems** that continually learn and develop skills to tackle increasingly complex problems.

Interests AI Agents, Reinforcement Learning, Foundation Models, LLMs / VLMs.

WORK EXPERIENCE



Student Researcher at Google

Oct. 2026 - Present

Montreal, QC, Canada

- Research on reinforcement learning post-training of LLM agents.



Research Intern at Microsoft Research

Jun. 2026 - Oct. 2026

MSR Montreal — Montreal, QC, Canada

- Post-training coding LLMs with RL and online task synthesis.
- Released FrogNano, a 4B coding agent trained purely with RL on synthetic tasks.



Research Intern at Vmax

Jan. 2026 - June 2026

San Francisco, CA, US

- LLM post-training with RL, SFT and on-policy distillation for coding agents.
- Synthetic data generation at scale with LLMs for SWE-agents.



Research Intern at Ubisoft LaForge

Jun. 2023 - Sep. 2023

Montreal, QC, Canada

- Improving the convergence of intrinsic exploration algorithms with deep RL.
- Integration of AI agents in 3D games for human-like navigation.



Junior Data Scientist at HP Inc

Feb. 2021 - Aug. 2022

Barcelona, Spain

- Built an optimization framework and an interactive web app (Flask, PostgreSQL) for automated budget allocation.

PUBLICATIONS

2026

GlyphBench: A Playground for Language-Model Reinforcement Learning

Roger Creus Castanyer, Marc-Alexandre Côté, Matthew James Sargent, Augustine N. Mavor-Parker, Glen Berseth, Pablo Samuel Castro. *Under review*

FrogNano: Training a 4B Coding Agent via Online Task Synthesis

Minseon Kim, Zhengyan Shi, Emiliano Penaloza, Christopher Cui, **Roger Creus Castanyer**, et al. *Preprint*

Agentick: A Unified Benchmark for General Sequential Decision-Making Agents

Roger Creus Castanyer, Pablo Samuel Castro, Glen Berseth. **NeurIPS 2026 · Evaluations & Datasets Track**

ARM-FM: Automated Reward Machines via Foundation Models for Compositional Reinforcement Learning

Roger Creus Castanyer, Faisal Mohamed, Pablo Samuel Castro, Cyrus Neary, Glen Berseth. **ICLR 2026**

PopuLoRA: Co-Evolving LLM Populations for Reasoning Self-Play

Roger Creus Castanyer, Geoffrey Bradway, Lorenz Wolf, Maxwell Lin, Augustine N. Mavor-Parker, Matthew James Sargent. *Under review*

unix-ctf: Procedural Environments for Unix-Competence Reinforcement Learning

Geoffrey Bradway, **Roger Creus Castanyer**, Lorenz Wolf, Maxwell Lin, Matthew James Sargent, Augustine N. Mavor-Parker. *Under review*

Breaking the Solver Bottleneck: Training Task Generators at the Learnable Frontier

Lorenz Wolf, Connor Watts, **Roger Creus Castanyer**, Geoffrey Bradway, Maxwell Lin, Augustine N. Mavor-Parker, Matthew Daborn-Sargent. *Under review*

2025

Stable Gradients for Stable Learning at Scale in Deep Reinforcement Learning

Roger Creus Castanyer, Obando, J., Lu Li, Pierre-Luc Bacon, Glen Berseth, Aaron Courville, Pablo S. Castro. **NeurIPS 2025** **Spotlight**

What Matters for Maximizing Data Reuse In Value-based Deep Reinforcement Learning

Roger Creus Castanyer, Glen Berseth, Pablo Samuel Castro. **Finding the Frame Workshop, RLC 2025**

2024

Improving Intrinsic Exploration by Creating Stationary Objectives

Roger Creus Castanyer, Joshua Romoff, Glen Berseth. **ICLR 2024**

Surprise-Adaptive Intrinsic Motivation for Unsupervised Reinforcement Learning

Adriana Hugessen (*), **Roger Creus Castanyer** (*), Faisal Mohamed (*), Glen Berseth. **RLC 2024 · IMOL Workshop, NeurIPS 2023** **Oral**

RLeXplore: Accelerating Research in Intrinsically-Motivated Reinforcement Learning

Mingqi Yuan (*), **Roger Creus Castanyer** (*), Bo Li, X. Jin, Glen Berseth, W. Zeng. **TMLR 2024**

2021

Unsupervised Skill-Discovery and Skill-Learning in Minecraft

Juanjo Nieto, **Roger Creus Castanyer**, Xavier Giro-i-Nieto. **URL Workshop, ICML 2021**

Integration of Convolutional Neural Networks in Mobile Applications

Roger Creus Castanyer, Xavier Franch, Silverio Martinez. **AI Engineering Workshop, ICSE 2021**

Which Design Decisions in AI-enabled Mobile Applications Contribute to Greener AI?

Roger Creus Castanyer, Xavier Franch, Silverio Martinez. **Empirical Software Engineering (EMSE) 2021**

SCHOLARSHIPS

- 2026** **Doctoral Research Scholarship (B2)** — Fonds de Recherche du Québec (FRQNT)
- 2025** **AI Research Scholarship** — University of Montreal
- 2024** **Academic Excellence Scholarship** — University of Montreal
- 2023** **End-of-studies Scholarship** — University of Montreal
- 2021** **Introduction to Research Scholarship (INIREC)** — Banco Santander, Spain

OTHER ACHIEVEMENTS

Contributor to Open-Source AI Software

Contributor to widely used AI repositories — Agentick, GlyphBench, CleanRL, RLLTE, RLeXplore, Wave-Defense RL, Blokus RL, Lux AI, and xgenius — all available on GitHub.

Content Creator in Artificial Intelligence

I run the YouTube channel [@whoami_AI](#) with educational AI content; some videos have reached 30k+ views.

Basketball Player — Fundació Privada Sagrada Família Claror | Sep. 2008 - Jun. 2020

Highest level played: the First Catalan League in the 2017–2018 and 2018–2019 seasons.

Basketball Coach — Fundació Privada Sagrada Família Claror | Sep. 2014 - Jun. 2022

Coached and mentored youth teams (ages 8–16) competing at a high level in multiple tournaments.
